RELIABILITY-AWARE CLINICAL TRIAGE

Framework for Uncertainty Quantification in Predictive Healthcare

Executive Overview

MedTech startups face what I call the trust Gap. While AI models can generate predictions, they rarely signal when they are working with ambiguous or incomplete clinical data. This brief presents a Split-Conformal Prediction framework that addresses this gap by providing statistically rigorous confidence sets for ICU triage decisions ensuring AI-driven recommendations come with mathematical safety guarantees.

The Problem: The "Black Box" Liability in Clinical AI

In clinical settings, an overconfident false negative is not just a poor metric, but it is a patient safety incident. Most machine learning models share three critical weaknesses:

- No uncertainty signal. They lack a formal mechanism to communicate when they don't know an answer.
- Probability drift. Performance degrades unpredictably when deployed in real-world environments.
- Regulatory vulnerability. They struggle to meet emerging FDA standards for AI transparency in clinical decision support.

The Solution: Conformal Inference Wrapper

Using a MIMIC-IV-ED inspired data architecture, I implemented a conformal wrapper that transforms point-estimate probabilities into nested prediction sets. This approach addresses the core reliability challenges while remaining practical for deployment.

Key Technical Pillars:

- Distribution-Free Coverage
  - Unlike Bayesian methods, this framework makes no parametric assumptions about the underlying data. This matters when working with messy, real-world EHR data where distribution assumptions rarely hold.
- Calibration-First Logic
  - A dedicated calibration set defines a non-conformity threshold ($\hat{q}$) that ensures the model achieves a pre-specified error rate (typically 5%). This calibration step is what enables the mathematical safety guarantee.
- Human-in-the-Loop Routing
  - The system automatically flags ambiguous cases, those where the confidence set contains multiple possible outcomes, for immediate clinical review. This preserves automation benefits while maintaining safety.

Methodology & Architecture

The pipeline utilizes the Split-Conformal Prediction algorithm as formalized by Angelopoulos & Bates (2021). By leveraging distribution-free uncertainty quantification, the system provides a mathematical guarantee of coverage that remains robust across the high-variance data characteristic of clinical environments.

- Dataset: Simulated EHR schema (Triage Acuity, SpO2, Heart Rate, Prior Admissions).
- Algorithm: XGBoost / Logistic Regression with a Split-Conformal wrapper.
- Evaluation: Empirical Coverage vs. Set Size efficiency.

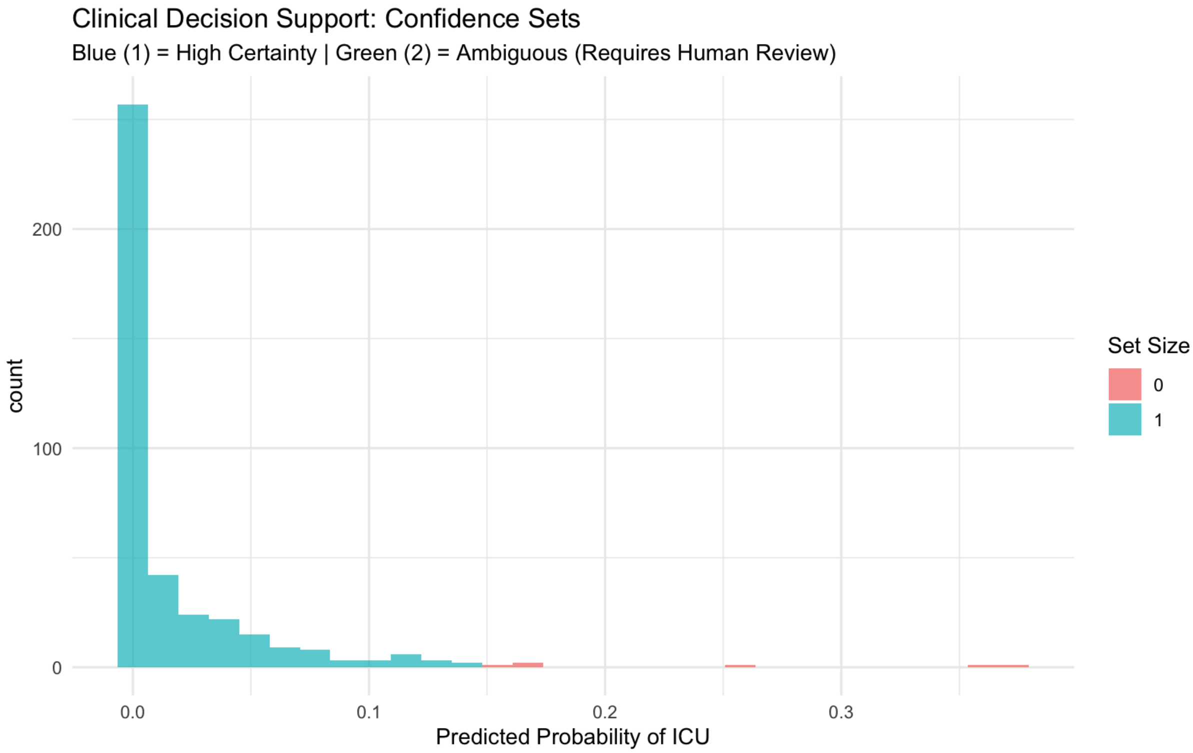

Figure 1. Distribution of prediction certainty. Blue bars indicate "High-Confidence" single-class predictions. Green bars highlight "Ambiguous" cases where the model recommends human intervention to maintain a 95% safety guarantee.

Strategic Business Impact

- De-Risking the Regulatory Path
  - The framework provides the statistical traceability that FDA and EMA increasingly require for AI-based Clinical Decision Support systems. This documentation becomes part of the regulatory submission package.
- Operational Scalability
  - By enabling "filtered automation," the system can handle roughly 80% of routine triage cases with near-zero error, allowing clinicians to focus their expertise on the 20% of complex, high-stakes cases.
- Reduced R&D
  - Early identification of out-of-distribution data reduces the cost of validating model predictions in live clinical trials. Failed predictions are caught before they reach expensive validation stages.

About the Researcher

Vani Patel is a Master's candidate in **Biostatistics & Health Data Science** at the University of Pittsburgh School of Public Health (Graduating Dec 2026).

- **Multidisciplinary Foundation:** Combines an undergraduate background in **Bioinformatics and Business** from the University of Waterloo with advanced training in machine learning and statistical inference.
- **Strategic Mindset:** Brings leadership and consulting experience to the startup space, prioritizing data solutions that align with clinical milestones and commercial ROI.
- **Current Thesis Focus:** Specialized research in **Prediction-Powered Inference (PPI)** and **Conformal Prediction** to solve the "Black Box" problem in digital health and medtech, bridging the gap between high-performance ML and rigorous statistical validity.

Full Technical Documentation & Code: Github

References & Inspiration

Angelopoulos, A. N., & Bates, S. (2021). *A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification.* [arXiv:2107.07511].

Concept adapted from the Conformal Triage framework (Angelopoulos et al.), originally applied to image classification and clinical decision support.